JASP Student Developers

Introduction to R-Coding for JASP

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Contents

1. The Main Analysis Function
2. Output Tables
   - Coding challenges

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The Main Analysis Function

```
BinomialTest <- function(jaspResults, dataset, options) {
```

- This is the function you assign to the analysis in the Description.qml file
- Arguments:
  - jaspResults
    - Contains results from the analysis for the JASP output
  - dataset
  - options
    - Contains interface options selected by the user

Dataset & Options

What do these actually look like?

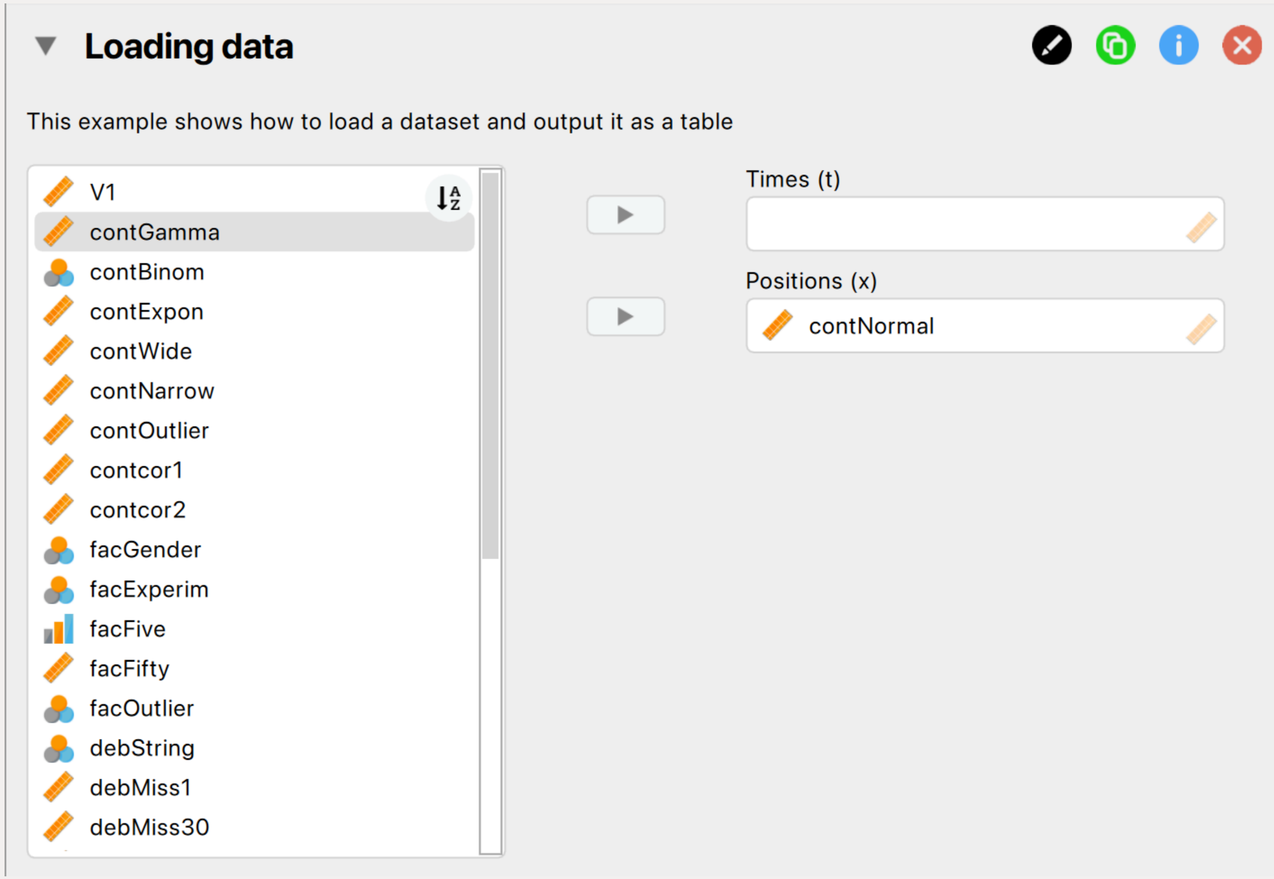
```
VariablesForm
{
  AvailableVariablesList { name: "allVariables" }

  AssignedVariablesList  {
    name: "ts"
    label: qsTr("Times (t)")
    info: qsTr("This info entry adds documentation to the (i) icon in the analysis file. E.g.,
    Specify variable containing the time.")
    singleVariable: true
    allowedColumns: ["scale"]
  }

  AssignedVariablesList  {
    name: "xs"
    label: qsTr("Positions (x)")
    info: qsTr("This info entry adds documentation to the (i) icon in the analysis file. E.g.,
    Specify variable containing the positions.")
    singleVariable: true
    allowedColumns: ["scale"]
  }
}
```

Dataset & Options

What do these actually look like?



Dataset & Options in R

What do these actually look like?

options

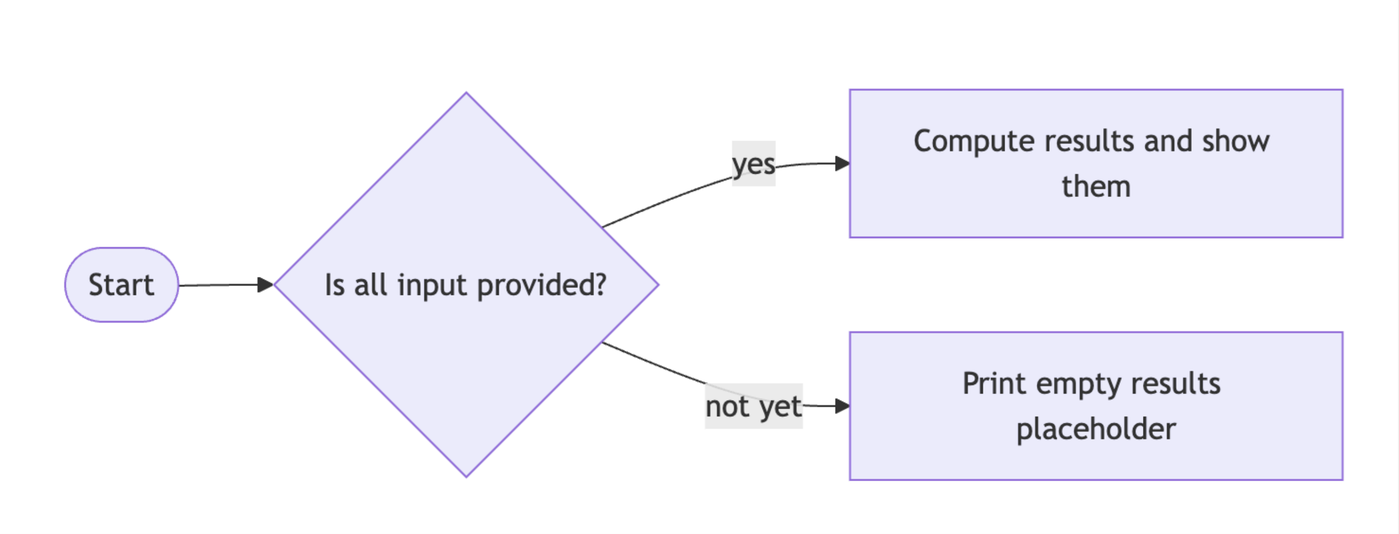
options	list [11]	List of length 11
.meta	list [3]	List of length 3
checkedFactor	character [1]	''
nominalOrOrdinalVa...	character [1]	''
nominalOrOrdinalVa...	character [1]	'unknown'
plotHeight	integer [1]	320
plotWidth	integer [1]	480
ts	character [1]	''
ts.types	list [0]	List of length 0
values	list [0]	List of length 0
xs	character [1]	'JaspColumn_15_Encoded'
xs.types	character [1]	'scale'

dataset

	JaspColumn_15_Encoded
1	0.99122228
2	-1.95334497
3	1.57630164
4	0.94793341
5	-0.53255250
6	-1.36359763
7	-0.65497938
8	-0.75144595
9	0.31018152
10	-0.67527073

Writing the Main R-Function

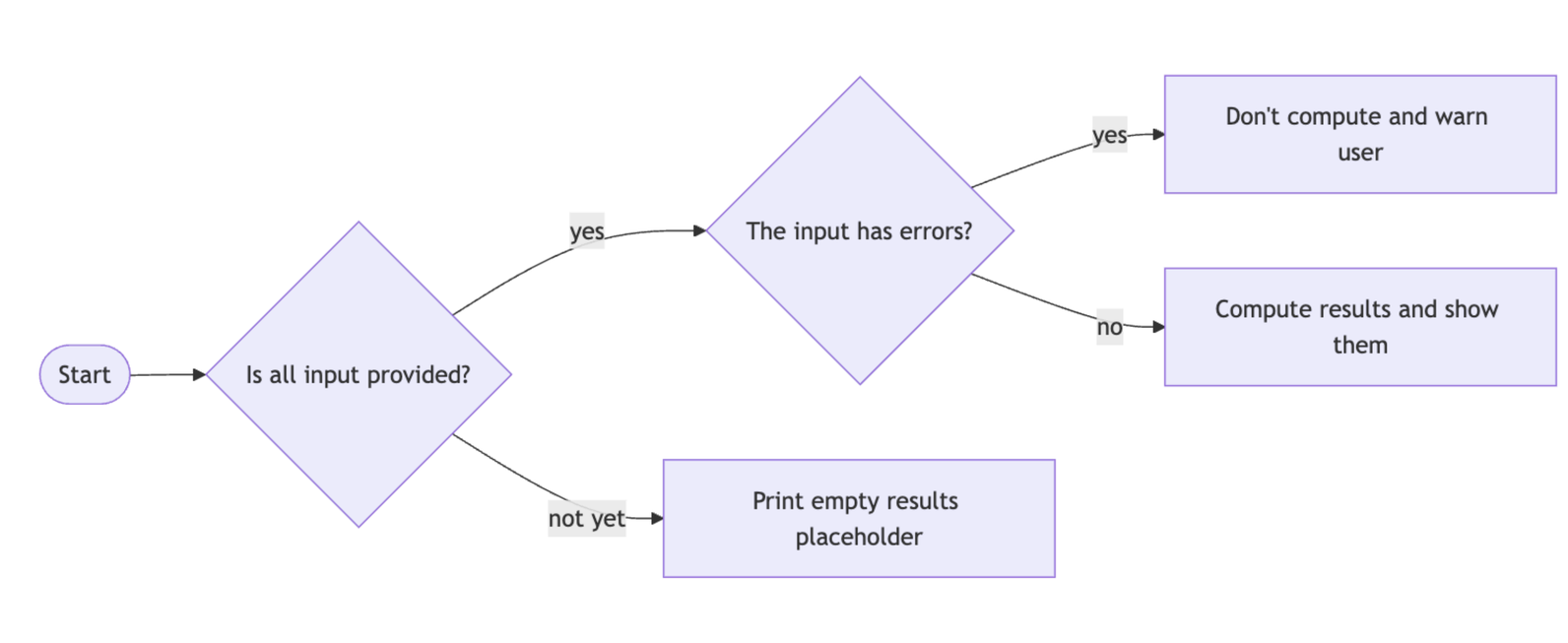
Check if results can be computed



More information here.

Writing the Main R-Function

Check for errors



More information here.

Contents

1. The Main Analysis Function
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Writing the Main R-Function

Output tables

1. Create table
2. Adding table to the output
3. Filling the table with results

Output Table

1. Creating the table

- Usually you have a separate function that handles this
- The `createJaspTable()` function from the `JaspBase` package creates the table object
- You can give it a title that will be displayed above the table in JASP

Output Table

1. Creating the table

```
# Output tables (each calls its own results function)
.ttestIndependentMainTable(  jaspResults, dataset, options, ready, type)
.ttestIndependentNormalTable(jaspResults, dataset, options, ready, type)
```

```
.ttestIndependentMainTable <- function(jaspResults, dataset, options, ready, type) {
  if (!is.null(jaspResults[["ttest"]]))
    return()

  optionsList <- .ttestOptionsList(options, type)

  # Create table
  ttest <- createJaspTable(title = gettext("Independent Samples T-Test"))
```

Output Table

1.1 Dependencies

- Dependencies = analysis options (from the interface) causing object to recompute
- Specified using `someTableName$dependOn()`

```
ttest$dependOn(c("effectSize", "effectSizeCi", "dependent",
                 "effectSizeCiLevel", "student",
                 "meanDifference", "meanDifferenceCi",
                 "meanDifferenceCiLevel", "alternative",
                 "vovkSellke", "naAction", "group", "effectSizeType",
                 "welch", "mannWhitneyU", "equalityOfVariancesTest", "equalityOfVariancesTestType"))
```

Output Table

1.2 Adding columns to the table

```
ttest$addColumnInfo(name = testStat, type = "number",  title = testStatName)
ttest$addColumnInfo(name = "df",     type = dfType,    title = gettext("df"))
ttest$addColumnInfo(name = "p",      type = "pvalue",  title = gettext("p"))
```

- You can add columns using `someTableName$addColumnInfo()`
- The addColumnInfo requires some arguments:
  - name: The name of the column (important for filling the table later)
  - type: Type of output that will be displayed in the column (string, number, integer, pvalue)
  - title: Column header displayed in JASP
- Note: There are other (optional) arguments.

Output Table

2. Adding table to output

- When you're done creating the table, you have to add it to the jaspResults

```
jaspResults[["ttest"]] <- ttest
```

- Here you usually add an if-statement:
  - If results are NOT ready to be computed, **`return()`**
  - Else you want to fill the table with results

```
if (!ready)
  return() # If not ready, do nothing
```

Challenge

Add an empty table to the JASP output

- Fork the jaspModuleTemplate or sync your fork
- Add an empty output table to the Loading data analysis
  - For example, add a table that will contain the mean of a variable
- Within your local clone of the module template, locate the *R* folder, and then open the *loadingData.R* file
- You do not need to fill the table with any output yet!

Output Table

3. Filling the table with results

- For filling the table, you have two options

1. `someTableName$addRows()`
   - Expects list with names of the entries equal to the names of the columns that you defined when creating the table
2. `someTableName$setData()`
   - Can be used when you would like to add all rows at once
   - Accepts data.frames among others

Output Table

3. Filling the table with results

```
for (variable in options$variables) {

  <calculate our results with stats::binom.test>

  binomialTable$addRows(list(variable = variable,
                             level         = level,
                             counts        = counts,
                             total         = nObs,
                             proportion    = prop,
                             p             = p,
                             VovkSellkeMPR = vovkSellkeMPR,
                             lowerCI       = lowerCI,
                             upperCI       = upperCI))
}
```

Challenge

Fill your table with results

- Fill the table you created previously with results (e.g., the mean of an input variable)
- More details about R-coding for JASP can be found in the documentation.